Informatica® Cloud Data Integration

Hadoop Files V2 Connector

Informatica Cloud Data Integration Hadoop Files V2 Connector
October 2025

© Copyright Informatica LLC 2019, 2026

This software and documentation are provided only under a separate license agreement containing restrictions on use and disclosure. No part of this document may be reproduced or transmitted in any form, by any means (electronic, photocopying, recording or otherwise) without prior consent of Informatica LLC.

U.S. GOVERNMENT RIGHTS Programs, software, databases, and related documentation and technical data delivered to U.S. Government customers are "commercial computer software" or "commercial technical data" pursuant to the applicable Federal Acquisition Regulation and agency-specific supplemental regulations. As such, the use, duplication, disclosure, modification, and adaptation is subject to the restrictions and license terms set forth in the applicable Government contract, and, to the extent applicable by the terms of the Government contract, the additional rights set forth in FAR 52.227-19, Commercial Computer Software License.

Informatica, the Informatica logo, Informatica Cloud, and PowerCenter are trademarks or registered trademarks of Informatica LLC in the United States and many jurisdictions throughout the world. A current list of Informatica trademarks is available on the web at https://www.informatica.com/trademarks.html. Other company and product names may be trade names or trademarks of their respective owners.

Portions of this software and/or documentation are subject to copyright held by third parties. Required third party notices are included with the product.

See patents at https://www.informatica.com/legal/patents.html.

DISCLAIMER: Informatica LLC provides this documentation "as is" without warranty of any kind, either express or implied, including, but not limited to, the implied warranties of noninfringement, merchantability, or use for a particular purpose. Informatica LLC does not warrant that this software or documentation is error free. The information provided in this software or documentation may include technical inaccuracies or typographical errors. The information in this software and documentation is subject to change at any time without notice.

NOTICES

This Informatica product (the "Software") includes certain drivers (the "DataDirect Drivers") from DataDirect Technologies, an operating company of Progress Software Corporation ("DataDirect") which are subject to the following terms and conditions:

1. THE DATADIRECT DRIVERS ARE PROVIDED "AS IS" WITHOUT WARRANTY OF ANY KIND, EITHER EXPRESSED OR IMPLIED, INCLUDING BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY, FITNESS FOR A PARTICULAR PURPOSE AND NON-INFRINGEMENT.
2. IN NO EVENT WILL DATADIRECT OR ITS THIRD PARTY SUPPLIERS BE LIABLE TO THE END-USER CUSTOMER FOR ANY DIRECT, INDIRECT, INCIDENTAL, SPECIAL, CONSEQUENTIAL OR OTHER DAMAGES ARISING OUT OF THE USE OF THE ODBC DRIVERS, WHETHER OR NOT INFORMED OF THE POSSIBILITIES OF DAMAGES IN ADVANCE. THESE LIMITATIONS APPLY TO ALL CAUSES OF ACTION, INCLUDING, WITHOUT LIMITATION, BREACH OF CONTRACT, BREACH OF WARRANTY, NEGLIGENCE, STRICT LIABILITY, MISREPRESENTATION AND OTHER TORTS.

The information in this documentation is subject to change without notice. If you find any problems in this documentation, report them to us at infa_documentation@informatica.com.

Informatica products are warranted according to the terms and conditions of the agreements under which they are provided. INFORMATICA PROVIDES THE INFORMATION IN THIS DOCUMENT "AS IS" WITHOUT WARRANTY OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING WITHOUT ANY WARRANTIES OF MERCHANTABILITY, FITNESS FOR A PARTICULAR PURPOSE AND ANY WARRANTY OR CONDITION OF NON-INFRINGEMENT.

Publication Date: 2026-07-10

Table of Contents

Preface 5

Informatica Resources 5

- Informatica Documentation 5
- IDMC web site 5
- IDMC Communities 5
- IDMC Marketplace 5
- Data Integration connector documentation 6
- Informatica Knowledge Base 6
- IDMC Trust Center 6
- Informatica Global Customer Support 6

Chapter 1: Introduction to Hadoop Files V2 Connector 7

Hadoop Files V2 Connector example 7

Hadoop Files V2 Connector assets 7

Hadoop Files V2 connector administration 8

- Access Kerberos-enabled Hadoop cluster 8
- Access the non-Kerberos enabled Hadoop cluster 8

Hadoop distributions for Hadoop Files V2 Connector 9

Chapter 2: Hadoop Files V2 connections 10

Hadoop Files V2 connection properties 10

Hadoop Files V2 Connector rules and guidelines 12

Chapter 3: Mappings and tasks with Hadoop Files V2 Connector 13

Hadoop Files V2 sources in mappings 13

Hadoop Files V2 targets in mappings 15

- Specifying a target 19
- Target time stamps 19
- Writing to multiple target objects 20

Hadoop Files V2 lookups in mappings 20

Hadoop Files formatting options 23

- Read and write flat file formats 23
- Rules and guidelines for file formats 26

Hadoop Files V2 compression formats 27

- Data compression for Hadoop Files V2 sources and targets 27
- Configuring LZO compression format 28

File partitioning to read from complex files 28

Running a mapping on Azure HDInsights Kerberos cluster with WASB storage 29

Reading and writing data from column names with hash characters 29

Rules and guidelines for Hadoop Files V2 objects in mappings 30

Appendix A: Hadoop Files V2 data type reference. 31

Flat file data types and transformation data types. 31

Avro complex file data types and transformation data types. 32

Parquet complex file data types and transformation data types. 33

JSON complex file data types and transformation data types. 34

Rules and guidelines for data types. 35

Index. 36

Preface

Use the *Hadoop Files V2 Connector* to learn how to read from or write to Hadoop Distributed File System by using Cloud Data Integration. Learn to create a connection and develop and run mappings, mapping tasks, and dynamic mapping tasks in Cloud Data Integration.

Informatica Resources

Informatica provides you with a range of product resources through the Informatica Network and other online portals. Use the resources to get the most from your Informatica products and solutions and to learn from other Informatica users and subject matter experts.

Informatica Documentation

Use the Informatica Documentation Portal to explore an extensive library of documentation for current and recent product releases. To explore the Documentation Portal, visit https://docs.informatica.com.

If you have questions, comments, or ideas about the product documentation, contact the Informatica Documentation team at infa_documentation@informatica.com.

IDMC web site

You can access the IDMC web site at http://www.informatica.com/cloud. This site contains information about Informatica Cloud integration services.

IDMC Communities

Use the IDMC Community to discuss and resolve technical issues. You can also find technical tips, documentation updates, and answers to frequently asked questions.

Access the IDMC Community at:

https://network.informatica.com/community/informatica-network/products/cloud-integration

Developers can learn more and share tips at the Cloud Developer community:

https://network.informatica.com/community/informatica-network/products/cloud-integration/cloud-developers

IDMC Marketplace

Visit the Informatica Marketplace to try and buy Data Integration Connectors, templates, and mapplets:

https://marketplace.informatica.com/

Data Integration connector documentation

You can access documentation for Data Integration Connectors at the Documentation Portal. To explore the Documentation Portal, visit https://docs.informatica.com.

Informatica Knowledge Base

Use the Informatica Knowledge Base to find product resources such as how-to articles, best practices, video tutorials, and answers to frequently asked questions.

To search the Knowledge Base, visit https://search.informatica.com. If you have questions, comments, or ideas about the Knowledge Base, contact the Informatica Knowledge Base team at KB_Feedback@informatica.com.

IDMC Trust Center

The IDMC Trust Center provides information about Informatica security policies and real-time system availability.

You can access the trust center at https://www.informatica.com/trust-center.html.

Subscribe to the IDMC Trust Center to receive upgrade, maintenance, and incident notifications. The Informatica Intelligent Cloud Services Status page displays the production status of all the Informatica cloud products. All maintenance updates are posted to this page, and during an outage, it will have the most current information. To ensure you are notified of updates and outages, you can subscribe to receive updates for a single component or all IDMC components. Subscribing to all components is the best way to be certain you never miss an update.

To subscribe, on the Informatica Intelligent Cloud Services Status page, click **SUBSCRIBE TO UPDATES**. You can choose to receive notifications sent as emails, SMS text messages, webhooks, RSS feeds, or any combination of the four.

Informatica Global Customer Support

You can contact a Global Support Center through the Informatica Network or by telephone.

To find online support resources on the Informatica Network, click **Contact Support** in the IDMC Help menu to go to the **Cloud Support** page. The **Cloud Support** page includes system status information and community discussions. Log in to Informatica Network and click **Need Help** to find additional resources and to contact Informatica Global Customer Support through email.

The telephone numbers for Informatica Global Customer Support are available from the Informatica web site at https://www.informatica.com/services-and-training/support-services/contact-us.html.

CHAPTER 1

Introduction to Hadoop Files V2 Connector

You can use Hadoop Files V2 Connector to securely read data from and write data to flat or complex files on local system or to HDFS (Hadoop Distributed File System). You can read or write structured, semi-structured, and unstructured data.

You can create a Hadoop Files V2 connection and use the connection in mappings or mapping tasks. You can switch mappings to advanced mode to include transformations and functions that enable advanced functionality. Create a mapping task to process data based on the data flow logic defined in a mapping or integration template.

Hadoop Files V2 Connector example

You are a data analyst in a large-scale social media enterprise. The enterprise stores departmental data in multiple formats such as PDF and JSON. A large amount of data is stored in the local file system of multiple computers. You need to consolidate this large amount of data and analyze it for business-critical decisions in near future.

You can consolidate data in the Hadoop environment and run various analytics tools to analyze the data quickly. You can use Hadoop Files V2 Connector to read data from complex files in multiple formats from the local file systems and write data to HDFS. After you write data to HDFS, use tools such as Hadoop Analytics to analyze the data.

Hadoop Files V2 Connector assets

Create assets in Data Integration to integrate data using Hadoop Files V2 Connector.

When you use Hadoop Files V2 Connector, you can include the following Data Integration assets:

- Mapping
- Mapping task

For more information about configuring assets and transformations, see *Mappings*, *Transformations*, and *Tasks* in the Data Integration documentation.

Hadoop Files V2 connector administration

Before you use Hadoop Files V2 Connector in tasks, an administrator must verify the following prerequisites:

- Use Hadoop Files V2 Connector on the Linux 64-bit operating system.
- Before you connect to the Hadoop cluster, enable all the required ports for the data node, name node, KMS, and Zookeeper services.

Access Kerberos-enabled Hadoop cluster

Configure the `/etc/hosts` file and copy the Kerberos configuration file for HDFS instances that use Kerberos authentication.

1. Open the `/etc/hosts` file located in the `/etc` directory on the Secure Agent machine on Linux.
2. To configure the Secure Agent to work with the Kerberos Key Distribution Center (KDC), make an entry of the KDC hosts in the `/etc/hosts` file.
3. Copy the `krb5.conf` configuration file from the `/etc` directory in the Hadoop cluster node to either of the following directories available within your Secure Agent installation:
   - `<Secure Agent installation directory>\jdk\jre\lib\security`
   - `<Secure Agent installation directory>\jdk8\jre\lib\security`
4. If there is a jdk directory within the `<Secure agent installation directory>\apps` folder, navigate to the following directory and copy the `krb5.conf` configuration file from the `/etc` directory in the Hadoop cluster node to the following directory within your Secure Agent installation:
   `<Secure agent installation directory>\apps\jdk\zulu8<latest_version>\jre\lib\security`
5. If the cluster is SSL enabled, import the certificate alias file to either of the following directories available within your Secure Agent installation:
   - `<Secure Agent installation directory>\jdk\jre\lib\security\cacerts`
   - `<Secure Agent installation directory>\jdk8\jre\lib\security\cacerts`
6. If there is a jdk directory within the `<Secure agent installation directory>\apps` folder, navigate to the following directory and import the certificate alias file to the following directory within your Secure Agent installation:
   `<Secure agent installation directory>\apps\jdk\zulu8<latest_version>\jre\lib\security\cacerts`
7. Restart the Secure Agent.

Access the non-Kerberos enabled Hadoop cluster

If you have a Cloudera, Amazon EMR, Hortonworks, or Microsoft HDInsight instance that does not use Kerberos authentication and runs in a Hadoop cluster environment, perform the following steps:

1. Copy the `conf` configuration files from the `/etc/hadoop/conf` directory in the Hadoop cluster node and place it in the Secure Agent location with full permission.

Hadoop distributions for Hadoop Files V2 Connector

You can use the following distribution versions for Hadoop Files V2 Connector:

- Cloudera CDH 6.1
- Amazon EMR 5.20, 6.3, and 7.12
- Azure HDInsight 4.0
- Hortonworks HDP 3.1
- Cloudera CDP 7.1 private cloud
- Cloudera CDW 7.2 public cloud

CHAPTER 2

Hadoop Files V2 connections

You can use a Hadoop Files V2 connection in mappings and mapping tasks.

Create a Hadoop Files V2 connection so that the agent can read from or write to simple and complex files in a local file system or HDFS. You can create a Hadoop Files V2 connection in the **Connections** page. Use the connection when you create a mapping or a mapping task.

Hadoop Files V2 connection properties

When you set up a Hadoop Files V2 connection, you must configure the connection properties.

The following table describes the Hadoop Files V2 connection properties:

Connection property	Description
Connection Name	Name of the Hadoop Files V2 connection.
Description	Description of the connection. The description cannot exceed 765 characters.
Use Secret Vault	Stores sensitive credentials for this connection in the secrets manager that is configured for your organization. This property appears only if secrets manager is set up for your organization. When you enable the secret vault in the connection, you can select which credentials that the Secure Agent retrieves from the secrets manager. If you don't enable this option, the credentials are stored in the repository or on a local Secure Agent, depending on how your organization is configured. **Note:** If you're using this connection to apply data access policies through pushdown or proxy services, you cannot use the Secret Vault configuration option. For information about how to configure and use a secrets manager, see "Secrets manager configuration" in the Administrator help.
Runtime Environment	The name of the runtime environment where you want to run the tasks. Select a Secure Agent or elastic runtime environment. For more information about how to configure and use the runtime environments, see *Runtime Environments* in the Administrator help.
User Name	Required to read data from HDFS. Enter a user name that has access to the single-node HDFS location to read data from or write data to.

Connection property	Description

NameNode URI	The URI to access HDFS.
	Use the following format to specify the name node URI in Cloudera, Amazon EMR, and Hortonworks distributions:
	`hdfs://<namenode>:<port>/`
	where,
	- `<namenode>` is the host name or IP address of the name node.
	- `<port>` is the port that the name node listens for remote procedure calls (RPC).
	To connect to the Hadoop cluster, specify the name node port `fs.defaultFS`.
	If the Hadoop cluster is configured for high availability, you must copy the `fs.defaultFS` value in the `core-site.xml` file and append `/` to specify the name node URI.
	For example, the following snippet shows the `fs.defaultFS` value in a sample `core-site.xml` file:
	`<property>`
	`<name>fs.defaultFS</name>`
	`<value>hdfs://nameservice1</value>`
	`<source>core-site.xml</source>`
	`</property>`
	In the above snippet, the `fs.defaultFS` value is
	`hdfs://nameservice1`
	and the corresponding name node URI is
	`hdfs://nameservice1/`
	Note: Specify either the name node URI or the local path. Do not specify the name node URI if you want to read data from or write data to a local file system path.

Local Path	A local file system path to read and write data. Read the following conditions to specify the local path: - You must enter **NA** in local path if you specify the name node URI. If the local path does not contain **NA**, the name node URI does not work. - If you specify the name node URI and local path, the local path takes the preference. The connection uses the local path to run all tasks. - If you leave the local path blank, the agent configures the root directory (/) in the connection. The connection uses the local path to run all tasks. - If the file or directory is in the local system, enter the fully qualified path of the file or directory. For example, `/user/testdir` specifies the location of a directory in the local system. Default value for Local Path is NA.
Configuration Files Path	The directory that contains the Hadoop configuration files. **Note:** Copy the core-site.xml, hdfs-site.xml, and hive-site.xmlfrom the Hadoop cluster and add them to a folder on the Linux Box.
Keytab File	The file that contains encrypted keys and Kerberos principals to authenticate the machine.
Principal Name	Users assigned to the superuser privilege can perform all the tasks that a user with the administrator privilege can perform.
Impersonation Username	You can enable different users to run mappings in a Hadoop cluster that uses Kerberos authentication or connect to sources and targets that use Kerberos authentication. To enable different users to run mappings or connect to big data sources and targets, you must configure user impersonation.

Note: When you read from or write to remote files, the **NameNode URI** and **Configuration Files Path** fields are mandatory. When you read from or write to local files, you require only the **Local Path** field.

Hadoop Files V2 Connector rules and guidelines

Consider the following rules and guidelines when you create a Hadoop Files V2 connection:

- Ensure that you restart the Secure Agent if you update the following connection properties:
  - **Impersonation Username**
  - **Keytab File**
  - **Principal Name**
  - **NameNode URI**
  - **Configuration Files Path**
- Do not use a colon (:) and double forward slash (//) characters in the **File path** field of a complex file source object.

CHAPTER 3

Mappings and tasks with Hadoop Files V2 Connector

When you create a mapping, you can configure a Source or Target transformation to represent a Hadoop Files V2 object.

You can specify a single flat or complex file as source and target objects.

Select the flat or complex file in the source and target object when you want to perform a read or write operation to a single flat or complex file. If you want to perform read or write operations to multiple files in a directory and its sub directories, specify a directory as the source object.

To specify a directory as a source object, enter the **File Path** value in the advanced source properties from where you want to read the files. To specify a directory as a target object to store the written files in the Hadoop file system, enter the **File Directory** value in the advanced target properties.

You can also configure a Lookup transformation in the mapping to lookup data in the flat file format.

After you configure a mapping, deploy the mapping in a mapping task. If you parameterize the connection or object in a mapping, you must specify the parameterized values when you create the mapping task.

Hadoop Files V2 sources in mappings

To read data from a flat or complex file, configure a Hadoop Files V2 object as the Source transformation in a mapping. You can configure a Source transformation to represent a single flat or complex file source.

Specify the name and description of the Hadoop Files V2 source. Configure the source and advanced properties for the source object.

The following table describes the Hadoop Files V2 source properties that you can configure in a Source transformation:

Source Property	Description
Connection	Name of the source connection or create a connection parameter.
Source Type	Type of source object. Select Single or Parameter as the source type.
Object	Select the source object from which you want to read data. Though selecting a source object is mandatory, the agent ignores this object. The agent processes the source object specified in File Path in advanced source properties.

Source Property	Description
Format	File format of the source object. You can select from the following file format types: - None - Flat file - Avro - Parquet - JSON Default is **None**. If you select **None** as the format type, the Secure Agent reads data in binary format. In advanced mode, **None** is not applicable. **Note:** You cannot read data from ORC and Apache Iceberg file formats even though they are listed in the **Formatting Options**.
Parameter	The parameter for the source object. Create or select the parameter for the source object. **Note:** The parameter property appears if you select parameter as the source type.

The following table describes the Hadoop Files V2 source advanced properties that you can configure in a Source transformation:

Advanced Property	Description
File path	Mandatory. Location of the file or directory from which you want to read data. Maximum length is 255 characters. If the path is a directory, all the files in the directory must have the same file format. If the file or directory is in HDFS, enter the path without the node URI. For example, /user/lib/testdir specifies the location of a directory in HDFS. The path must not contain more than 512 characters. If the file or directory is in the local system, enter the fully qualified path. For example, /user/testdir specifies the location of a directory in the local system.
File Format	Mandatory. Name and format of the file from which you want to read data. Specify the value in the following format: `<filename>.<format>` For example, `customer.avro`
Allow Wildcard Characters	Indicates whether you want to use wildcard characters for the source directory name or the source file name. If you select this option, you can use asterisk (*) wildcard character for the source directory name or the source file name in the **File path** field.
Allow Incremental Read	Indicates whether you want to read only the newly added files or files that have changed since the last time the mapping task ran.
Allow Recursive Read	Indicates whether you want to use wildcard characters to read complex files of the Parquet, Avro, or JSON formats recursively from the specified folder and its subfolders and files. You can use the wildcard character as part of the file or directory. For example, you can use a wildcard character to recursively read data from the following folders: - **/myfolder*/**. Returns all files within any folder or subfolder that has a pattern myfolder in the path name. - **/myfolder*/*.csv**. Returns all .csv files within any folder or subfolder that has a pattern myfolder in the path name. - **/myfolder*/ and file name is abc***. Returns all files that have a pattern abc within any folder or subfolder that has a pattern myfolder in the path name.

Advanced Property	Description
File Format	Specifies a file format of a complex file source. Select one of the following options: - Binary - Custom Input - Sequence File Format Default is Binary.
Input Format	The class name for files of the input file format. If you select input file format in the **File Format** field, you must specify the fully qualified class name implementing the `InputFormat` interface. To read files that use the Avro format, use the following input format: `com.informatica.avro.AvroToXML`
Input Format Parameters	Parameters for the input format class. Enter name-value pairs separated with a semicolon. Enclose the parameter name and value within double quotes. For example, use the following syntax: `"param1"="value1";"param2"="value2"`
Compression Format	Compression format of the source files. Select one of the following options: - None - Auto - DEFLATE - gzip - bzip2 - Lzo - Snappy - Custom
Custom Compression Codec	Required if you use custom compression format. Specify the fully qualified class name implementing the `CompressionCodec` interface.
isSchemaValidationSupported	Determines whether you want to validate data against a predefined schema to ensure the data structure and types conform to expected formats. Select this option to validate the incoming data. Do not select this option to read without any schema validation checks.
Tracing Level	Sets the amount of detail that appears in the log file. You can choose terse, normal, verbose initialization, or verbose data. Default is normal.

Hadoop Files V2 targets in mappings

To write data to a Hadoop Files V2, configure a Hadoop Files V2 object as the Target transformation in a mapping. You can configure a Target transformation to represent a single Hadoop Files V2 target.

When you use an Hadoop Files V2 target object, select an Hadoop Files object as the target.

The following table describes the Hadoop Files V2 target properties that you can configure in a Target transformation:

Target Property	Description
Connection	Name of the target connection or create a connection parameter.
Object Type	Type of target object. Select Single Object or Parameter.
Object	Select the file to which you want to write data. You can select an existing object or create an object at run time. When you specify a file name in the advanced target properties, the agent ignores the selected object and considers the values specified in the **File Name** in the advanced target properties.
Create New at Runtime	Creates a complex file target object at runtime. Enter a name for the target object and map the source fields that you want to use. By default, all source fields are mapped. You can use parameters defined in a parameter file in the target name.
Format	File format of the target object. You can select from the following file format types: - None - Flat file - Avro - Parquet - JSON Default is **None**. If you select **None** as the format type, the Secure Agent writes data in binary format. In advanced mode, **None** is not applicable. When you write binary data to a Hadoop Files V2 target created at run time, ensure that the incoming field that contains the binary data is named **data.** If the field name differs, use field rules or an Expression transformation to change the field name to **data**. **Note:** You cannot write data to ORC and Apache Iceberg file formats even though they are listed in the **Formatting Options**.
Parameter	The parameter for the target object. Create or select the parameter for the target object. **Note:** The parameter property appears if you select parameter as the target type.
Operation	Select the target operation. You can use only the insert operation.

The following table describes the Hadoop Files V2 target advanced properties that you can configure in a Target transformation:

Advanced Property	Description
File Directory	Optional. The directory location of one or more output files. Maximum length is 255 characters. If you do not specify a directory location, the output files are created at the location specified in the connection. If the directory is in HDFS, enter the path without the node URI. For example, /user/lib/testdir specifies the location of a directory in HDFS. The path must not contain more than 512 characters. If the file or directory is in the local system, enter the fully qualified path. For example, /user/testdir specifies the location of a directory in the local system.
File Name	Optional. The name of the output file. Enter the file name for your output file in the following format: `<file name>.<file extension>` For example, abc.txt. If you don't enter a file name in this field, the agent considers the file name of the target object selected from the Object field.
Generate Unique File Name	Generates a unique file name by appending unique identifiers to the file name. If you entered a file name, <file name>.<file extension>, in the **File Name** field, the Secure Agent generates the output file name in the following format: `<file name>_<partition index>_<hex of time in milliseconds>_<file sequence number>.<file extension>` For example, if you enter abc.txt in the **File Name** field, the Secure Agent generates the output file named abc_1_190fea05bdd_0001.txt. Where, - file name is abc - partition index is 1 - hex of time in milliseconds is 190fea05bdd (The current time in milliseconds converted into a hex string) - file sequence number is 0001 - file extension is .txt Default is enabled. Disable this option to generate the output file using the exact file name specified in the **File Name** field. For example, if you entered abc.txt as the file name, the Secure Agent generates the output file name without appending any unique identifiers. If you don't specify a file name in the **File Name** field, unique identifiers are appended to the target object selected from the Object field.
Overwrite Target	Indicates whether the Secure Agent first deletes the target data before writing data. If you select this option, the Secure Agent deletes all files that match the file name specified in the **File Name** field. For example, if you enter abc.txt in the **File Name** field with the **Overwrite Target** option enabled, the Secure Agent deletes all files named abc*.txt. If you do not select this option, the Secure Agent creates a new file in the target and writes the data to that file.
File Format	Specifies a file format of a complex file source. Select one of the following options: - Binary - Custom Input - Sequence File Format Default is Binary.

Advanced Property	Description
Output Format	The class name for files of the output format. If you select Output Format in the **File Format** field, you must specify the fully qualified class name implementing the `OutputFormat` interface.
Output Key Class	The class name for the output key. If you select Output Format in the **File Format** field, you must specify the fully qualified class name for the output key. You can specify one of the following output key classes: - BytesWritable - Text - LongWritable - IntWritable **Note:** Hadoop Files V2 generates the key in ascending order.
Output Value Class	The class name for the output value. If you select Output Format in the **File Format** field, you must specify the fully qualified class name for the output value. You can use any custom writable class that Hadoop supports. Determine the output value class based on the type of data that you want to write. **Note:** When you use custom output formats, the value part of the data that is streamed to the complex file data object write operation must be in a serialized form.
Compression Format	Compression format of the source files. Select one of the following options: - None - Auto - DEFLATE - gzip - bzip2 - LZO - Snappy - Custom
Custom Compression Codec	Required if you use custom compression format. Specify the fully qualified class name implementing the `CompressionCodec` interface.
Sequence File Compression Type	Optional. The compression format for sequence files. Select one of the following options: - None - Record - Block
streamRolloverSize	The maximum size a file can reach before it is segmented into a new file. The rollover size manages large data streams by breaking them into smaller, more manageable parts.
streamRollovertime	The time interval after which a new file is created to write the data, regardless of the file size. This time interval helps to manage files by time and ensures data segmentation over fixed time periods.
Forward Rejected Rows	Determines whether the transformation passes rejected rows to the next transformation or drops rejected rows. By default, the mapping task forwards rejected rows to the next transformation.

Specifying a target

You can use an existing target or create a target to hold the results of a mapping. If you choose to create the target, the agent creates the target when you run the task.

To specify the target properties, follow these steps:

1. Select the Target transformation in the mapping.
2. On the **Incoming Fields** tab, configure field rules to specify the fields to include in the target.
3. To specify the target, click the **Target** tab.
4. Select the target connection.
5. For the object type, choose **Single Object** or **Parameter**.
6. In the **Object** field, select a target object or enter a parameter file path.
7. To choose a target object, click **Select**. You can select an existing target object or create a new target object at run time and specify the object name.
   - To create a target file at run time, enter the name for the target file including the extension. For example, `Accounts.csv`.
   - If you want the file name to include a time stamp, click **Handle Special Characters** and add special characters to the file name. For example, add the special characters shown here to include all the time stamp information: `Accounts_%d%m%y%T.csv`.
   - If you want the folder name to include a time stamp, click **Handle Special Characters** and add the folder name separated with a back slash (\) followed by the file name. For example, `%Y%m%d \Target_filename_%m.csv`.
8. Click **Formatting Options** if you want to configure the formatting options for the file, and click **OK**.
9. Specify the advanced properties for the target, if needed.

Target time stamps

When you create a target at run time in a mapping, you can append time stamp information to the file name to show when the file is created.

When you specify the file name for the target file, include special characters based on Linux STRFTIME function formats that the mapping task uses to include time stamp information in the file name. The time stamp is based on the organization's time zone.

The following table describes some common STRFTIME function formats that you might use in a mapping or mapping task:

Special Characters	Description
%d	Day as a two-decimal number, with a range of 01-31.
%m	Month as a two-decimal number, with a range of 01-12.
%y	Year as a two-decimal number without the century, with range of 00-99.
%Y	Year including the century, for example 2015.
%T	Applicable only to flat files. Time in 24-hour notation, equivalent to %H:%:M:%S.
%H	Hour in 24-hour clock notation, with a range of 00-24.

Special Characters	Description
%I	Hour in 12-hour clock notation, with a range of 01-12.
%M	Minute as a decimal, with a range of 00-59.
%S	Second as a decimal, with a range of 00-60.
%p	Either AM or PM.

Writing to multiple target objects

When you import target objects, the Secure Agent appends a FilePath field to the imported target object. When you map the FilePath field in the target object to an incoming field, the Secure Agent creates the folder structure and the target files based on the FilePath field. For example:

Syntax:

```
<tgt_FilePath_folder>/<tgt_FilePath=incoming_value_folder>/part_file
```

Sample:

```
emp_tgt.parquet/emp_tgt.parquet=128000/part-0000-e9ca8-6af-efd43-455c-8709.c000.parquet
```

The FilePath field is applicable to the following file formats:

- Avro
- Parquet

Consider the following guidelines when using the target FilePath field in mappings:

- Do not map the source object FilePath field to the target object FilePath field. If you map the FilePath field in the target object to an incoming field, the Secure Agent does not create directory structure as expected.
- When you map a date type incoming field to the FilePath field in the target object, the Secure Agent creates a nested folder structure based on the incoming date value for target objects.

Hadoop Files V2 lookups in mappings

You can configure a connected lookup operation to lookup data in flat files. To lookup data from a flat file, configure a Hadoop Files V2 object as a Lookup transformation in a mapping.

You can configure a Lookup transformation to represent a single flat file source. Specify the name and description of the Hadoop Files V2 lookup. Configure the lookup and advanced properties for the lookup object.

The following table describes the Hadoop Files V2 lookup properties that you can configure in a Lookup transformation:

Source Property	Description
Connection	Name of the source connection or create a connection parameter.
Source Type	Type of source object. Select Single or Parameter as the source type.
Lookup Object	Select the source object from which you want to read data. Though selecting a source object is mandatory, the agent ignores this object. The agent processes the source object specified in File Path in advanced source properties.
Parameter	The parameter for the source object. Create or select the parameter for the source object. **Note:** The parameter property appears if you select parameter as the source type.
Format	File format of the source object. You can select from the following file format types: - None - Avro - Parquet - JSON Default is **None**. If you select **None** as the format type, the Secure Agent reads data in binary format. In advanced mode, **None** is not applicable. **Note:** You cannot read data from ORC and Apache Iceberg file formats even though they are listed in the **Formatting Options**.
Multiple Matches	

The following table describes the Hadoop Files V2 advanced properties that you can configure in a Lookup transformation:

Advanced Property	Description
File path	Mandatory. Location of the file or directory from which you want to read data. Maximum length is 255 characters. If the path is a directory, all the files in the directory must have the same file format. If the file or directory is in HDFS, enter the path without the node URI. For example, /user/lib/testdir specifies the location of a directory in HDFS. The path must not contain more than 512 characters. If the file or directory is in the local system, enter the fully qualified path. For example, /user/testdir specifies the location of a directory in the local system.
File Format	Mandatory. Name and format of the file from which you want to read data. Specify the value in the following format: `<filename>.<format>` For example, `customer.avro`
Allow Wildcard Characters	Indicates whether you want to use wildcard characters for the source directory name or the source file name. If you select this option, you can use asterisk (*) wildcard character for the source directory name or the source file name in the **File path** field.
Allow Incremental Read	Indicates whether you want to read only the newly added files or files that have changed since the last time the mapping task ran.

Advanced Property	Description
Allow Recursive Read	Indicates whether you want to use wildcard characters to read complex files of the Parquet, Avro, or JSON formats recursively from the specified folder and its subfolders and files. You can use the wildcard character as part of the file or directory. For example, you can use a wildcard character to recursively read data from the following folders: - **/myfolder*/**. Returns all files within any folder or subfolder that has a pattern myfolder in the path name. - **/myfolder*/*.csv**. Returns all .csv files within any folder or subfolder that has a pattern myfolder in the path name. - **/myfolder*/ and file name is abc***. Returns all files that have a pattern abc within any folder or subfolder that has a pattern myfolder in the path name.
File Format	Specifies a file format of a complex file source. Select one of the following options: - Binary - Custom Input - Sequence File Format Default is Binary.
Input Format	The class name for files of the input file format. If you select input file format in the **File Format** field, you must specify the fully qualified class name implementing the `InputFormat` interface. To read files that use the Avro format, use the following input format: `com.informatica.avro.AvroToXML`
Input Format Parameters	Parameters for the input format class. Enter name-value pairs separated with a semicolon. Enclose the parameter name and value within double quotes. For example, use the following syntax: `"param1"="value1";"param2"="value2"`
Compression Format	Compression format of the source files. Select one of the following options: - None - Auto - DEFLATE - gzip - bzip2 - Lzo - Snappy - Custom
Custom Compression Codec	Required if you use custom compression format. Specify the fully qualified class name implementing the `CompressionCodec` interface.
isSchemaValidationSupported	Determines whether you want to validate data against a predefined schema to ensure the data structure and types conform to expected formats. Select this option to validate the incoming data. Do not select this option to read without any schema validation checks.
Tracing Level	Sets the amount of detail that appears in the log file. You can choose terse, normal, verbose initialization, or verbose data. Default is normal.

Hadoop Files formatting options

You can select the format of the Hadoop File and configure the formatting options when you read from or write to flat or complex files on the local system or to HDFS. You can use Avro, Parquet, JSON, and flat file format to read from or write data.

The following table describes the formatting options for Avro, Parquet, JSON, and delimited flat files in the Source and Target transformations:

Property	Description
Schema Source	The schema of the source or target file. Select one of the following options to specify a schema: - Read from data file. Imports the schema from a file in Hadoop Files. - Import from schema file. Imports the schema from a schema definition file in the agent machine.
Schema File	The schema definition file in the agent machine from where you want to upload the schema. You cannot upload a schema file when you create a target at runtime. The Schema File option appears if you select **Import from schema file**.

Read and write flat file formats

You can read from or write to delimited flat files. However, you cannot use Hadoop Files V2 Connector to read or write from fixed width flat files.

You need to enable staging before you can read or write to delimited flat files, When you set the staging property, Data Integration, by default, creates a flat file locally in a temporary folder to stage the data before it read from or writes to delimited flat files.

Enable staging to use flat file formats

You can set Data Integration to use staging for read, lookup, and write operations that involve data with delimited flat file formats.

If you do not set the staging property, Data Integration does not perform staging and fails to read from or write to a flat file. This property is specific to flat files and has no impact while processing other data formats.

Staging properties

You can set staging for read, lookup, and write operations.

You need to set the following staging property in the Secure Agent properties based on the operation that you want to configure:

Operation	Property to set
Read operation	INFA_DTM_RDR_STAGING_ENABLED_CONNECTORS
Lookup operation	INFA_DTM_LKP_STAGING_ENABLED_CONNECTORS
Write operation	INFA_DTM_STAGING_ENABLED_CONNECTORS

Set the staging property

Perform the following tasks to set the staging property for the Tomcat in the Secure Agent properties:

1. In Administrator, click **Runtime Environments**.
2. Edit the Secure Agent for which you want to set the property.
3. In the **System Configuration Details** section, select the **Service** as **Data Integration Server** and the type as **Tomcat**.
4. Set the value of the Tomcat property to the plugin ID of Hadoop Files V2 Connector.
   You can find the plugin ID in the manifest file located in the following directory: `<Secure Agent installation directory>/downloads/<Hadoop Files V2 package>/CCIManifest.txt`

When you run the mapping, a temporary flat file is created in the following directory in your machine based on the operation for which you set the property:

- Read operation: `C:\Windows\Temp\StagingReader\<Source_Name><data_.csv>`
- Lookup operation: `C:\Windows\Temp\StagingLookup\<Lookup_Name><data_.csv>`
- Write operation: `C:\Windows\Temp\snowflake\stage\<Snowflake_Target.txt>`

To use a temporary directory, add the following property and required path value to the JVMOptions of the Secure Agent: *"-Dhadoopfiles.tmpdir=/home/tmp/"*. The Secure Agent uses this designated location to create temporary files. If you do not provide a location, the Secure Agent uses the Java's temporary directory.

Data Integration logs the following messages in the session log if staging is done successfully through the flat file:

- Read operation: `Staging mode is enabled to read data.`
- Lookup operation: `Staging mode is enabled to read data.`
- Write operation: `The INFA_DTM_STAGING is successfully enabled to use the flat file to create local staging files.`

For rules and guidelines that applies for flat file formats, see Flat file formatson page 26.

Flat file formatting options

The following table describes the formatting options for flat files:

Property	Description
Flat File Type	The type of flat file. Select the Delimited option to reads a flat file that contains column delimiters. The fixed width option doesn't apply for Hadoop Files.
Delimiter	Character used to separate columns of data in a delimited flat file. You can set values as comma, tab, colon, semicolon, or others. You cannot set a tab as a delimiter directly in the **Delimiter** field. To set a tab as a delimiter, you must type the tab character in any text editor. Then, copy and paste the tab character in the **Delimiter** field.
EscapeChar	Character immediately preceding a column delimiter character embedded in an unquoted string, or immediately preceding the quote character in a quoted string data in a delimited flat file. When you write data to Hadoop Files and specify a qualifier, by default, the qualifier is considered as the escape character. Else, the character specified as the escape character is considered.

Property	Description
Qualifier	Quote character that defines the boundaries of data in a delimited flat file. You can set qualifier as single quote or double quote.
Qualifier Mode	The qualifier behavior when you write data to a delimited flat file. It determines whether you want data available in the staging file to be enclosed in double quotes. The following qualifier modes display: - MINIMAL - ALL - NON_NUMERIC - ALL_NON_NULL You need to select ALL to enclose all entries in double quotes. The rest of the options doesn't apply for Hadoop Files.
Disable escape character when a qualifier is set	Not applicable.
Code Page	Select the code page that the Secure Agent must use to read or write data to a delimited flat file. Select UTF-8 for mappings. Select one of the following options for mappings in advanced mode: - UTF-8 - MS Windows Latin 1 - Shift-JIS - ISO 8859-15 Latin 9 (Western European) - ISO 8859-3 Southeast European - ISO 8859-5 Cyrillic - ISO 8859-9 Latin 5 (Turkish) - IBM EBCDIC International Latin-1
Header Line Number	Not applicable.
First Data Row	Specify the line number from where you want the Secure Agent to read data in a delimited flat file. You must enter a value that is greater or equal to one. To read data from the header, the value of the **Header Line Number** and the **First Data Row** fields should be the same. Default is 1.
Target Header	Select whether you want to write data to a target that contains a header or without a header in the delimited flat file. You can select **With Header** or **Without Header** options. This property is not applicable when you read data from a Hadoop Files source.
Distribution Column	Not applicable.
Max Rows To Preview	Not applicable.
Row Delimiter	Not applicable.

Rules and guidelines for file formats

You must set the appropriate source and target properties when you select the file format types.

File path

Consider the following general guidelines when you specify the file path property:

- The **Object** field in the source properties from which you want to read data should be same as the source object specified in the **File path** field in advanced source properties.
- If you write data to a complex file target and the **File Path** field is mapped to any of the source fields, the file name in the **Object** field in the target properties must be different from the target **File Name** specified in the advanced target properties.

Create target

When you configure a Target transformation to create a new complex file target, consider the following guidelines:

- You can configure the format options and specify the file format type of the target object. If you select **None** as the format type, the Secure Agent writes data in binary format. Default is **None**. You cannot write data to ORC file format types even though it are listed in the **Formatting Options**.
- You cannot write data to a complex file object in Avro, Parquet, or JSON file format using the **Create Target** option, if one of the field names is FilePath in the source schema.

Flat file format

Consider the following guidelines when you enable the staging property to read from or write to delimited flat files:

- To read and write Null values as Null, add ReadNullAsNullCSVStaging property for the read operation and WriteNullAsNullCSVStaging for the write operation in the **Optional Connection Properties** in the Hadoop Files connection.
- You can configure only the following advanced properties in the Source transformation to read from delimited flat files:

  File path

  File pattern

  Allow recursive read

  Allow wildcard characters
- You can configure only the file directory, file name, and overwrite target fields in the advanced properties in the Target transformation to write to delimited flat files

JSON format

Consider the following guidelines for JSON files:

- You cannot read and write nested and multi-line indented JSON files.
  You can use the following JSON file structure to read data from and write data to a complex file object in JSON file format:
  - `{"Field Name1": "value", "Field Name2": "value", "Field Name3": "value",.....,"Field Name n": "value"}`
  - `[\{"Field Name1": "value", "Field Name2": "value", "Field Name3": "value",.....,"Field Name n": "value"},\{"Field Name1": "value", "Field Name2": "value", "Field Name3": "value",.....,"Field Name n": "value"},\{"Field Name1": "value", "Field Name2": "value", "Field Name3": "value",.....,"Field Name n": "value"}]`

Hadoop Files V2 compression formats

You can read or write compressed complex files, specify compression formats for the target objects, and decompress the output files.

You can use compression formats such as Gzip and DEFLATE.

The following table describes the complex file compression formats:

Compression Options	Description
None or empty	The file is not compressed.
Auto	The Data Integration Service detects the compression format of the file based on the file extension.
DEFLATE	The DEFLATE compression format that uses a combination of the LZ77 algorithm and Huffman coding.
Gzip	The GNU zip compression format that uses the DEFLATE algorithm.
Bzip2	The Bzip2 compression format that uses the Burrows–Wheeler algorithm.
LZO	The Lempel-Ziv-Oberhumer(LZO) compression format uses the LZO data-compression algorithm.
Snappy	The Snappy compression format uses the Snappy compression algorithm.

Data compression for Hadoop Files V2 sources and targets

You can decompress data when you read data from Hadoop Files V2 and compress the data when you write data to Hadoop Files V2.

Configure the compression format in the **Compression Format** option under the advanced source and target properties.

The following table lists the compression format that you use for the file formats:

File format	Compression format
Avro	None, Auto, Deflate, Snappy
JSON	None, Auto
Parquet	None, Auto, Gzip, LZO, Snappy
Binary	None, Auto, Bzip2, Deflate, Gzip, LZO, Snappy
Flat	None

Configuring LZO compression format

To write .jar files in the LZO compression format, you must copy the files for compression to the machine where the Secure Agent runs.

Perform the following steps to configure the Secure Agent for LZO compression:

1. Copy the LZO native binaries from the cluster to one of the following directories on the machine on which the Secure Agent runs:
   - `<agent-root>/downloads/package-<distribution>/package/<distribution name>/lib/native`
2. In the **Control Panel** Window, click **System and Security**.
3. In the **System and Security** Window, click **Advanced system settings**.
4. On the **Advanced** tab, select the **Environment Variables** button.
   The **Edit Environment Variables** dialog box appears.
5. Click **New** to add a new environment variable.
   The **New Environment Variables** dialog box appears.
6. Enter the value of the **Name** field as `LD_LIBARY_PATH`.
7. Enter the following path in the **Value** field:
   `<agent-root>/downloads/package-<distribution>/package/<distribution name>/infaLib/`
8. Restart the Secure Agent.

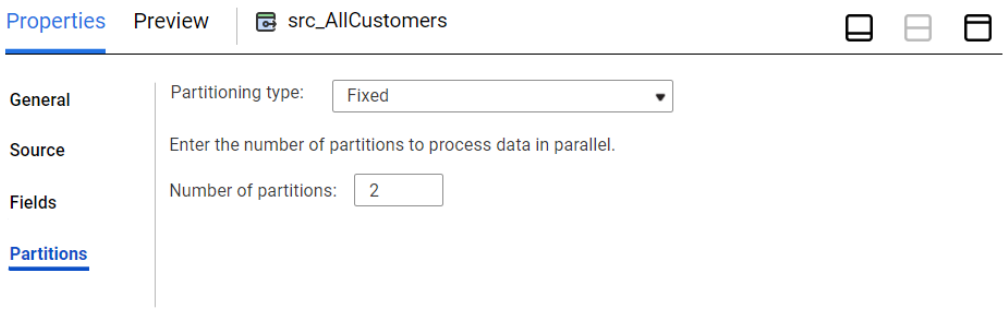
File partitioning to read from complex files

You can configure fixed partitioning in a mapping that reads data from complex files of the Parquet and Avro file formats.

Enable partitioning when you configure the Source transformation in the Mapping Designer.

On the **Partitions** tab for the Source transformation, you select fixed partitioning and enter the number of partitions based on the amount of data that you want to read.

The following image shows the configured partitioning:



When you run the mapping, the Secure Agent distributes the source data across the partitions to process parallely based on the number of partitions that you specify in the Source transformation. The maximum number of partitions must not exceed 64.

Running a mapping on Azure HDInsights Kerberos cluster with WASB storage

To read and process data from sources that use a Kerberos-enabled environment, you must configure the Kerberos configuration file, create user authentication artifacts, and configure Kerberos authentication properties for the Informatica domain.

To run a mapping for Hadoop Files V2 Connector using the Azure HDInsights with Windows Azure Storage Blob (WASB) kerberos cluster, perform the following steps:

1. Go to the `/usr/lib/python2.7/dist-packages/hdinsight_common/` directory on the Hadoop cluster node.
2. Run the following command to decrypt the account key:
   /decrypt.sh ENCRYPTED ACCOUNT KEY
3. Edit the `core-site.xml` file, in Agent conf location.
4. Replace the encrypted account key provided in the **fs.azure.account.key.STORAGE_ACCOUNT_NAME.blob.core.windows.net** property with the decrypted key, received as the output of the step #2.
5. Comment out the following properties to disable encryption and decryption of the account key:
   - **fs.azure.account.keyprovider.STORAGE_ACCOUNT_NAME.blob.core.windows.net**
   - **fs.azure.shellkeyprovider.script**
6. Save the `core-site.xml` file.
7. Copy the hdinsight_common folder from `/usr/lib/python2.7/dist-packages/hdinsight_common/` to the Secure Agent location.
8. Open the `core-site.xml` file in a browser to verify if the xml tags appear and ensure that there are no syntax issues.
9. Restart the Secure Agent.

Reading and writing data from column names with hash characters

To read and write data from columns that contain hash characters (#) in column names, you need to set the *export REPLACE_SPECIAL_CHARS* environment variable on the Secure Agent machine.

When you run a mapping that uses the Secure Agent configured with the *export REPLACE_SPECIAL_CHARS* environment variable, the Secure Agent replaces hash characters with underscore characters (_) in column names.

1. To configure the environment variable, stop the Secure Agent that is specified in the connection.
2. Set the following environment variable on the Secure Agent machine:
   export REPLACE_SPECIAL_CHARS=true
3. Restart the Secure Agent.
4. Map all columns in the field mapping and verify that underscore characters (_) replace hash characters in the displayed column names.
5. Run the mapping.

Rules and guidelines for Hadoop Files V2 objects in mappings

Consider the following rules and guidelines when you configure mappings:

- You can't preview the source and target objects in the mapping.
- To read or write data from columns that contain hash characters (#) in column names, set the *export REPLACE_SPECIAL_CHARS* environment variable on the Secure Agent machine. For more information, see "Reading and writing data from column names with hash characters" on page 29.

  Note: You can use hash characters (#) only in the Parquet file format.

APPENDIX A

Hadoop Files V2 data type reference

Data Integration uses complex files in mapping tasks with Hadoop Files V2 Connector.

Data Integration uses the following data types in mappings and mapping tasks with complex file:

Complex file native data types

Complex native data types appear in the Source and the Target transformations when you choose to edit metadata for the fields.

Transformation data types

Set of data types that appear in the transformations. They are internal data types based on ANSI SQL-92 generic data types, which the Secure Agent uses to move data across platforms. Transformation data types appear in all transformations in a mappings and mapping tasks.

When Data Integration reads source data, it converts the native data types to the comparable transformation data types before transforming the data. When Data Integration writes to a target, it converts the transformation data types to the comparable native data types.

The following table lists the complex file data types that Data Integration supports and the corresponding transformation data types:

Complex File Native Data Type	Transformation Data Type
Binary	Binary

Flat file data types and transformation data types

Flat file data types map to transformation data types that the Secure Agent uses to move data across platforms.

The following table lists the Hadoop Files flat data types that the Secure Agent supports and the corresponding transformation data types:

Flat File Data Type	Transformation Data Type	Description
BigInt	BigInt	-9,223,372,036,854,775,808 to 9,223,372,036,854,775,807 Precision of 19, scale of 0
Number	Decimal	Precision 1 to 28 digits, scale 0 to 28
Nstring	Text	1 to 104,857,600 characters Fixed-length or varying-length string.
STRING	String	1 to 104,857,600 characters Default precision is 256. You can increase the value up to 104857600 characters.

Note: You can read or write BigInt, Number, and Nstring data types only when you select **Import from Schema File** as the **Schema Source** in the file formatting options.

Avro complex file data types and transformation data types

Avro complex file data types map to transformation data types that the Secure Agent uses to move data across platforms.

The following table lists the Avro complex file data types that the Secure Agent supports and the corresponding transformation data types:

Avro Complex File Data Type	Transformation Data Type	Range and Description
Boolean	Integer	TRUE (1) or FALSE (0)
Bytes	Binary	Precision 4000
Double	Double	Precision 15
Float	Double	Precision 15
Int	Integer	-2,147,483,648 to 2,147,483,647 Precision 10, scale 0
Long	Bigint	-9,223,372,036,854,775,808 to 9,223,372,036,854,775,807 Precision 19, scale 0

Avro Complex File Data Type	Transformation Data Type	Range and Description
Null	Integer	-2,147,483,648 to 2,147,483,647 Precision 10, scale 0
String	String	-1 to 104,857,600 characters

Parquet complex file data types and transformation data types

Parquet complex file data types map to transformation data types that the Secure Agent uses to move data across platforms.

The following table lists the Parquet complex file data types that the Secure Agent supports and the corresponding transformation data types:

Parquet Complex File Data Type	Transformation Data Type	Range and Description
Binary	Binary	1 to 104,857,600 bytes.
Boolean	Integer	TRUE (1) or FALSE (0).
Date	Date/Time	January 1, 0001 to December 31, 9999.
Decimal	Decimal	Precision 1 to 28 digits, scale 0 to 28. **Note:** You cannot use decimal values with precision greater than 28.
Double	Double	Precision 15.
Float	Double	Precision 15.
Int32	Integer	-2,147,483,648 to +2,147,483,647.
Int64	Bigint	-9,223,372,036,854,775,808 to +9,223,372,036,854,775,807. 8-byte signed integer.
Int96	Date/Time	Jan1,0001 to Dec 31, 9999. Precision of 29, scale of 9.
String	String	-1 to 104,857,600 characters.

Parquet Complex File Data Type	Transformation Data Type	Range and Description
Time	Date/Time	Time of the day. Precision to microsecond.
Timestamp	Date/Time	January 1, 0001 00:00:00 to December 31, 9999 23:59:59.997. Precision to microsecond. **Note:** Precision to nanoseconds is not applicable.

The Parquet schema that you specify to read or write a Parquet file must be in smaller case. Case-sensitive schema is not applicable.

JSON complex file data types and transformation data types

JSON complex file data types map to transformation data types that the Secure Agent uses to move data across platforms.

The following table lists the JSON complex file data types that the Secure Agent supports and the corresponding transformation data types:

JSON Complex File Data Type	Transformation Data Type	Range and Description
Double	Double	Precision 15
Integer	Integer	-2,147,483,648 to 2,147,483,647 Precision of 10, scale of 0
Long	BigInt	-9,223,372,036,854,775,808 to 9,223,372,036,854,775,807 Precision 19, scale 0
String	String	-1 to 104,857,600 characters

Rules and guidelines for data types

Consider the following rules and guidelines for data types:

- Parquet data types support:
  - The following table shows the parquet version and the supported data types:

Parquet Version	Supported Data Types	Supported Distributions
2.1.0	Decimal	CDH 5.15
2.3.0	Date, Time_Millis, Timestamp_Millis	HDP 2.6, HDI 3.6
2.3.1	Date, Time_Millis, Time_Micros, Timestamp_Millis, Timestamp_Micros	CDH 6.1
2.4.0	Date, Time_Millis, Time_Micros, Timestamp_Millis, Timestamp_Micros	HDP 3.1, HDI 4.0

  - You can use the Decimal data types when you run a mapping in CDH 5.15, CDH 6.1, HDP 2.6, HDP 3.1, and HDI 4.0 distributions.

INDEX

A

administration 8

C

Cloud Application Integration community
  URL 5
Cloud Developer community
  URL 5
configuring
  lzo and snappy compression format 28
connection
  creating a 7
connections
  hadoop files 10
connections Hadoop Files V2 10
create target
  adding time stamps 19
  target file parameterization 19

D

data compression
  sources and targets 27
Data Integration community
  URL 5
data type reference
  overview 31

F

flat file
  staging data 23

H

hadoop files
  compression 27
  decompression 27
  target transformation 15
  targets in mappings 15
hadoop files V2
  mappings 13
Hadoop Files V2
  connection properties 10
  file formatting options 23
  rules and guidelines in mapping tasks 30
  specifying targets 19
Hadoop Files V2 Connector
  assets 7
  example 7
Hadoop Files V2 Connector *(continued)*
  overview 10

I

Informatica Global Customer Support
  contact information 6
Informatica Intelligent Cloud Services
  web site 5

J

JSON complex file data types
  transformation data types 34

M

maintenance outages 6
mappings
  hadoop files 13
  hadoop files target properties 15

O

ODBC connection
  rules and guidelines 12

P

Partitioning
  fixed partitioning for Hive sources 28

R

rules and guidelines
  Hadoop Files V2 file format types 26
Running a mapping
  running a mapping on Azure HDInsights with WASB Storage 29

S

staging data
  flat file 23
status
  Informatica Intelligent Cloud Services 6
system status 6

T

target
  FilePath field 20
target transformation
  hadoop files properties 15
targets
  hadoop files in mappings 15
trust site
  description 6

U

upgrade notifications 6

W

web site 5